

DIGITAL FORENSICS INVESTIGATION REPORT

Case Ref: EM-INV-26

Subject: Email Investigation

Target file: Nps -2010-emails

Examiner: DFI Charles

Date of Report: July 28, 2026

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EXECUTIVE SUMMARY

The **nps-2010-emails test disk** is a simple collection of **31 files**, each containing **one email address**. What makes it special is that every file stores its email address in a **different format**. Some formats are easy for computers to read, while others are more complicated.

This disk is used to **test whether computer programs can still find email addresses even when the text is saved in unusual or tricky ways**. It helps developers and digital-forensics teams make sure their tools work correctly on all kinds of files, not just the simple ones.

In plain terms:

It's a practice tool that checks if software can reliably find email addresses, no matter how the information is stored.

INCIDENT & CASE OVERVIEW

The **2010-NPS-Emails** disk image contains 31 uniquely encoded documents, each holding a different email address. Its inconsistent encoding schemes caused failures in multiple search-based tools, prompting the need for a formal incident case and project overview.

DFIR WORKFLOW STAGES

[1. Identification] → [2. Preservation] → [3. Acquisition] [4. Examination] → [5. Analysis] → [6. Documentation] → [7. Reporting] → [8. Presentation]

DIGITAL FORENSICS METHODOLOGY

The investigation followed strict ISO/IEC 27037 digital forensics standards covering the multi-stage lifecycle

1. **Identification:** Specifying media and digital evidence related to the incident.
2. **Preservation:** Maintaining evidence integrity through cryptographic hashing.
3. **Acquisition:** Creating bit-stream forensic images (E01).
4. **Examination:** Forensic extraction of active, system, and deleted files
5. **Analysis:** Translating raw technical artifacts into legally defensible findings.
6. **Documentation:** Continuous real-time record keeping of tools, paths, and procedures.
7. **Reporting:** Synthesis of findings into a structured, technical report
8. **Presentation:** Expert presentation of evidence to decision-makers

Primary Forensic Toolset : Autopsy 4.22.1 Forensic processing, ingest module parsing, artifact extraction, and deleted file recovery

EVIDENCE ACQUISITION & CHAIN OF CUSTODY

Digital Corpora

Producing the Digital Body

Wordpress search term

SEARCH WORDPRESS SITE

Corpus filename search

SEARCH CORPUS

AWS S3 Explorer

digitalcorpora / corpora / drives / nps-2010-emails

Hide folders?

English

50

entries per page

Search:

Object	Last Modified	Timestamp	Size
nps-2010-emails.E01	6 years ago	2020-11-21 21:18:46	507 KB
nps-2010-emails.strings	6 years ago	2020-11-21 21:18:47	85 KB
nps-2010-emails.txt	6 years ago	2020-11-21 21:18:47	2 KB
nps-2010-emails.zip	6 years ago	2020-11-21 21:18:48	465 KB

Showing 1 to 4 of 4 entries

Camera | Reports | File Dump | About

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Source file link:

https://digitalcorpora.s3.amazonaws.com/s3_browser.html#corpora/drives/nps-2010-emails/

Nps-2010-emails. E01 SHA256 Value:

sha256:c9ffd969954c2f9b9f97f459916c3d2e8755f596eda952c306ab3f9bc0d43bf1

INVENTIVE

Services Industries Learn Software Tools Resources About

Search...

Contact

SHA-256 of nps-2010-emails.E01:

c9ffd969954c2f9b9f97f459916c3d2e8755f596eda952c306ab3f9bc0d43bf1

Copy

Results Summary

Export TXT Export CSV Export JSON Clear Results

1

Total Hashes

0

Malicious

0

Known-Good

1

No Matches

No matches found. This hash isn't in any of the malware or known-good sources we check.

HASH	TYPE	STATUS	MALWARE NAME	THREAT LEVEL	DETECTION	FIRST SEEN	AC
c9ffd969954c2f9b9f97f459916c3d2e8755f596eda...	SHA-256	NO MATCH					

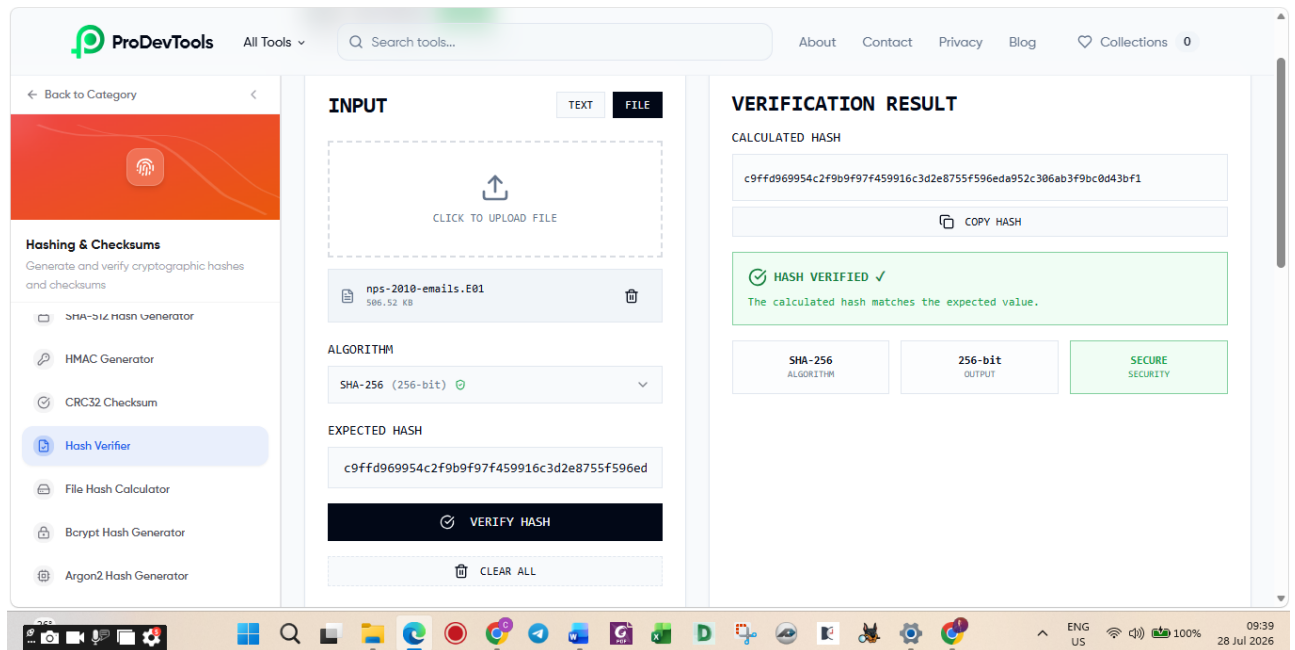
Suspicious Files on Your Network?

Our incident response team investigates malware, forensic analysis, and contains breaches.

https://inventivehq.com

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Nps-2010-emails. E01 SHA256 Value After verification:



sha256: c9ffd969954c2f9b9f97f459916c3d2e8755f596eda952c306ab3f9bc0d43bf1

Integrity Statement: Verification of the SHA-256 hash confirms that **Nps-2010-emails. E01** remained unaltered and bit-stream identical throughout the examination process.

EXAMINATION & TECHNICAL ANALYSIS

Email Data :

Digital Corpora

Sponsored by the AWS Open Data Sponsorship Program

About DigitalCorpora

Home

Corpora

nps-2010-emails

2010-nps-emails is a test disk that can be used for testing programs that find email addresses or perform string search.

The disk image consists of 30 different email addresses, each one stored in a different document with a different coding scheme.

Below are a list of the email addresses and their codings:

email address	Application (Encoding)
plain_text@textedit.com	Apple.TextEdit (UTF-8)
plain_text_pdf@textedit.com	Apple.TextEdit print-to-PDF (/FlateDecode)
rtf_text@textedit.com	Apple.TextEdit (RTF)
rtf_text_pdf@textedit.com	Apple.TextEdit print-to-PDF (/FlateDecode)
plain_utf16@textedit.com	Apple.TextEdit (UTF-16)
plain_utf16_pdf@textedit.com	Apple.TextEdit print-to-PDF (/FlateDecode)
pages@iwork09.com	Apple.Pages '09
pages_comment@iwork09.com	Apple.Pages (comment) '09
keynote@iwork09.com	Apple.Keynote '09
keynote_comment@iwork09.com	Apple.Keynote (comment) '09

Website Search

Search

Search the corpus:

SEARCH

Recent Posts

[CIRCL Forensics Exercises](#)

[Compiled bulk_extractor 2.0 ready for download](#)

Task View

Search

File Explorer

Edge

Chrome

Firefox

Discord

WhatsApp

Telegram

Skype

Zoom

Microsoft Word

Microsoft Excel

Microsoft PowerPoint

Microsoft OneNote

Microsoft Teams

Microsoft Outlook

Microsoft Access

Microsoft Publisher

Microsoft Project

Microsoft Visio

Microsoft SharePoint

Microsoft Dynamics

Microsoft Dynamics 365

Microsoft Dynamics CRM

Microsoft Dynamics NAV

Microsoft Dynamics SL

Microsoft Dynamics GP

Microsoft Dynamics AX

Microsoft Dynamics AX 2012

Microsoft Dynamics AX 2013

Microsoft Dynamics AX 2015

Microsoft Dynamics AX 2016

Microsoft Dynamics AX 2017

Microsoft Dynamics AX 2018

Microsoft Dynamics AX 2019

Microsoft Dynamics AX 2020

Microsoft Dynamics AX 2021

Microsoft Dynamics AX 2022

Microsoft Dynamics AX 2023

Microsoft Dynamics AX 2024

Microsoft Dynamics AX 2025

Microsoft Dynamics AX 2026

Microsoft Dynamics AX 2027

Microsoft Dynamics AX 2028

Microsoft Dynamics AX 2029

Microsoft Dynamics AX 2030

Microsoft Dynamics AX 2031

Microsoft Dynamics AX 2032

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Microsoft Dynamics AX 2089

Microsoft Dynamics AX 2090

Microsoft Dynamics AX 2091

Microsoft Dynamics AX 2092

Microsoft Dynamics AX 2093

Microsoft Dynamics AX 2094

Microsoft Dynamics AX 2095

Microsoft Dynamics AX 2096

Microsoft Dynamics AX 2097

Microsoft Dynamics AX 2098

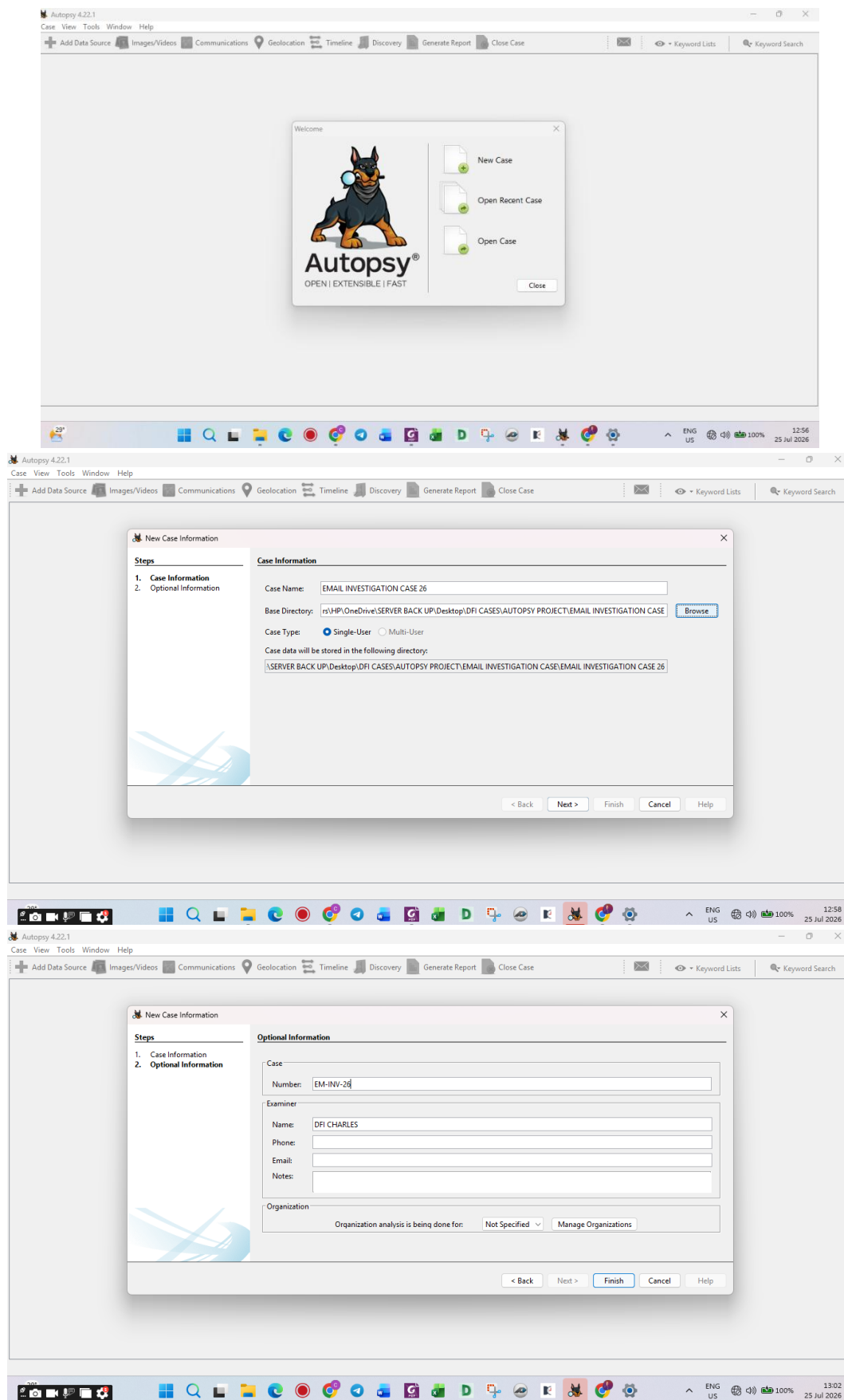
Microsoft Dynamics AX 2099

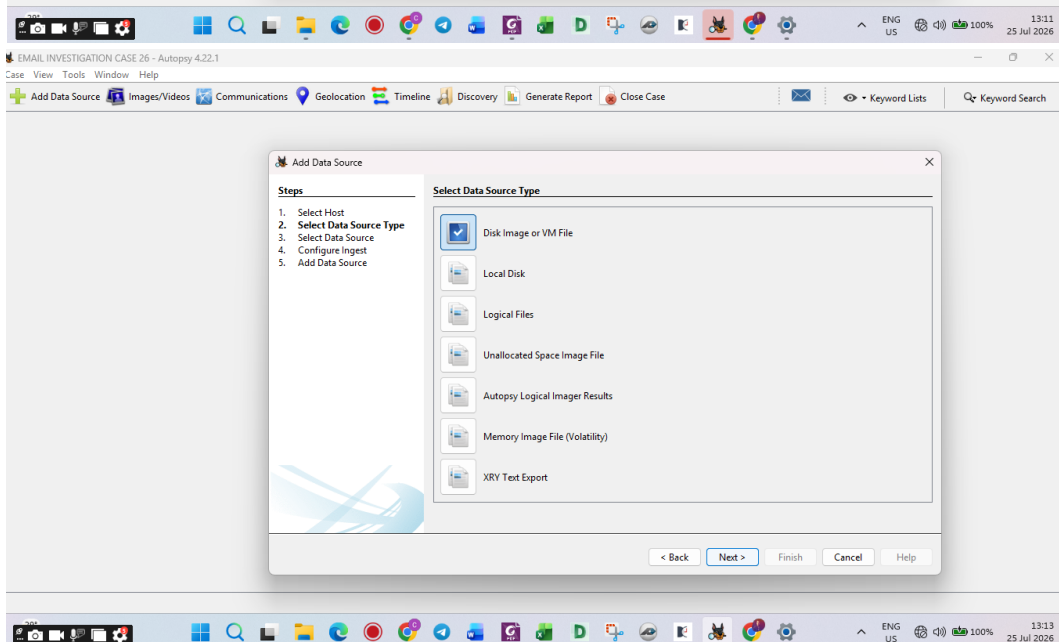
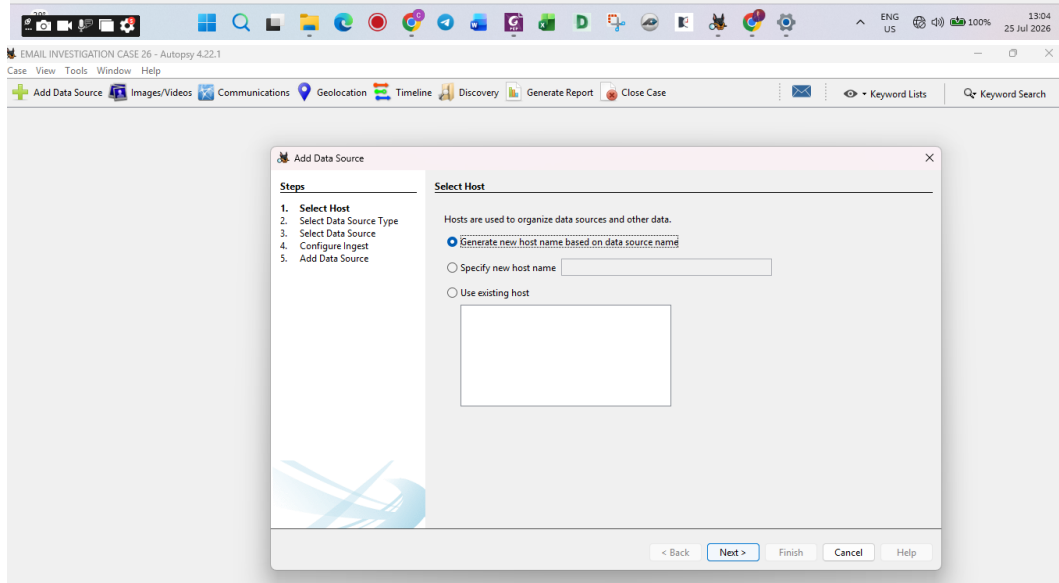
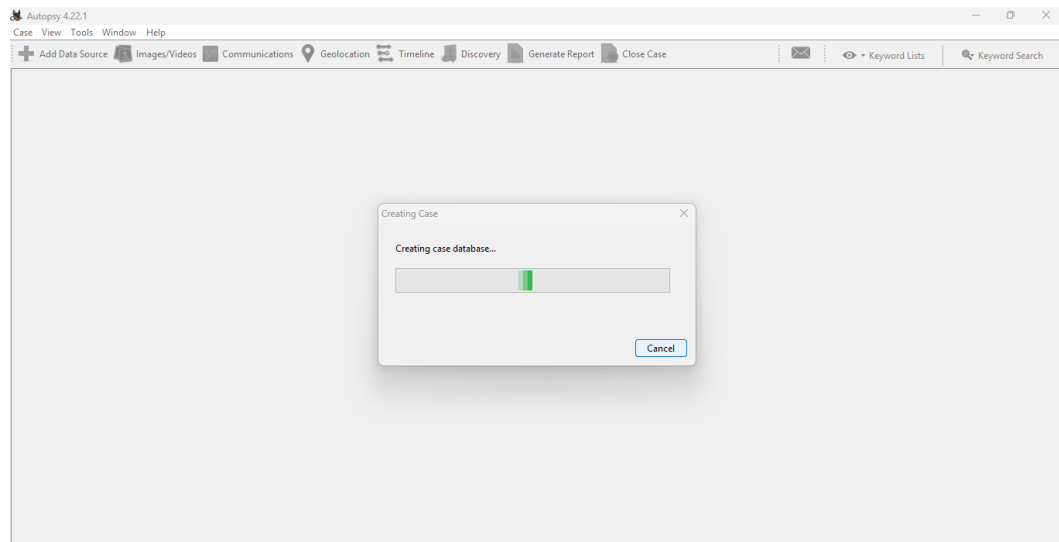
Microsoft Dynamics AX 2100

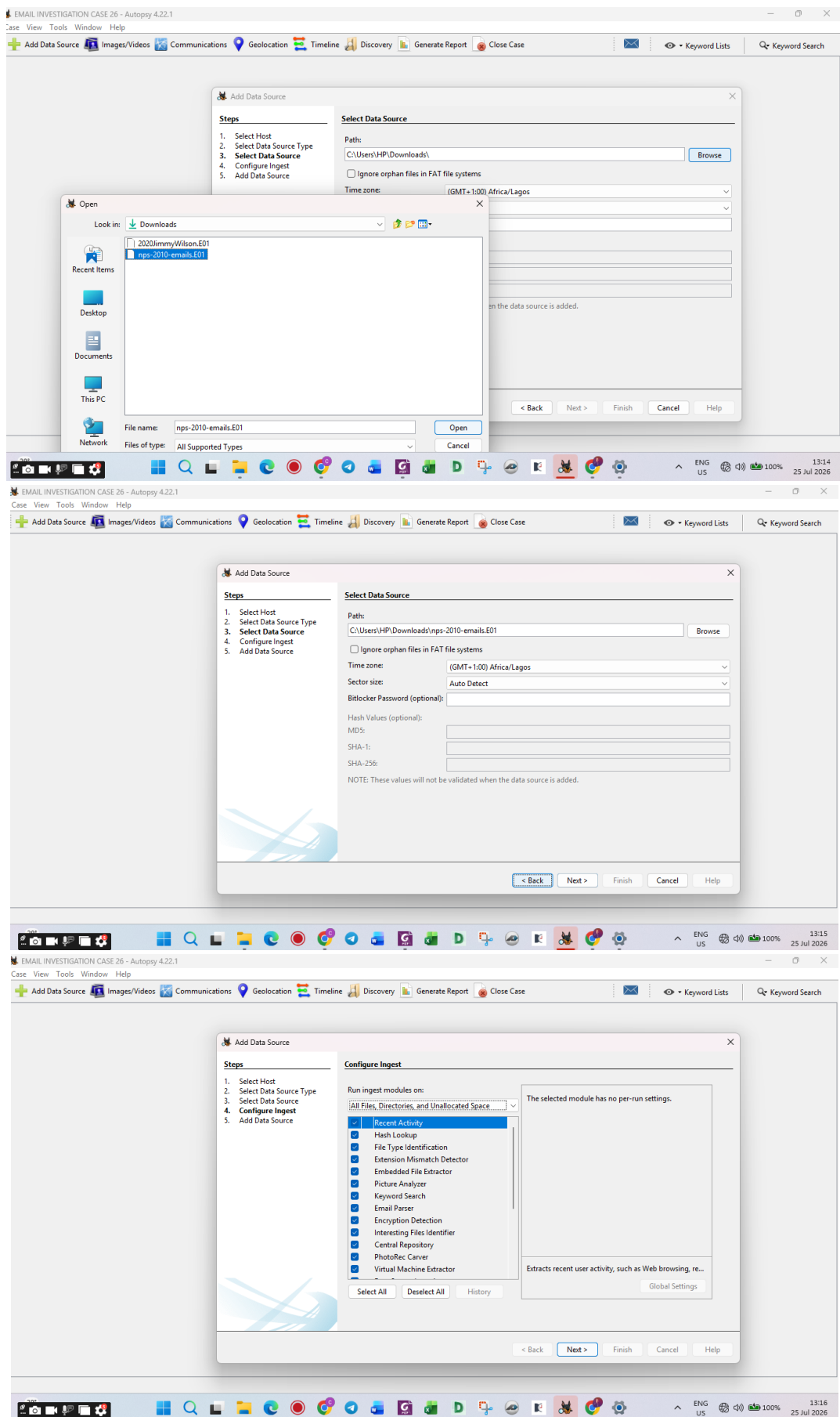
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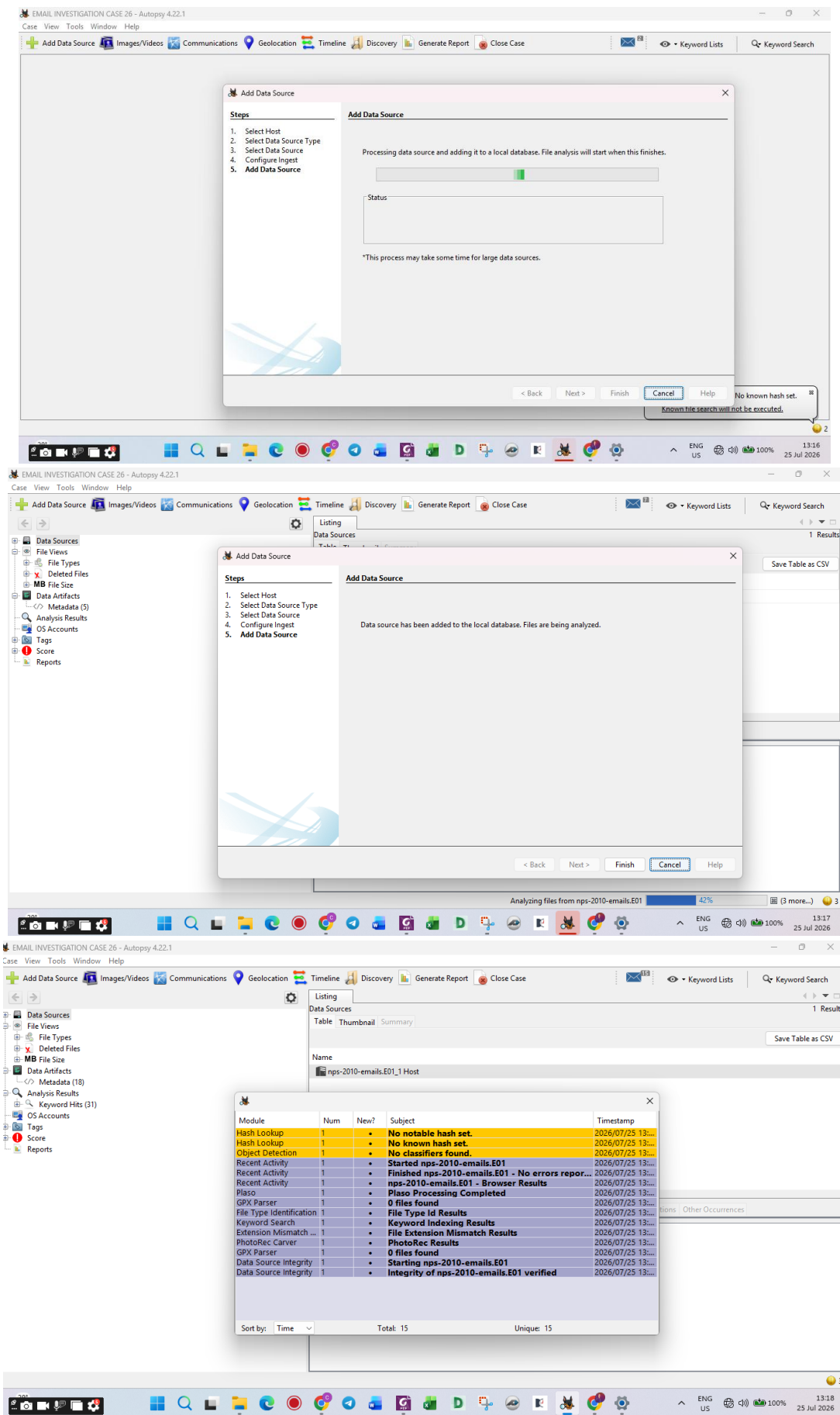
Evidence that there are over 30 emails to be recovered from this E01 file

System Artifacts & Recent Documents :



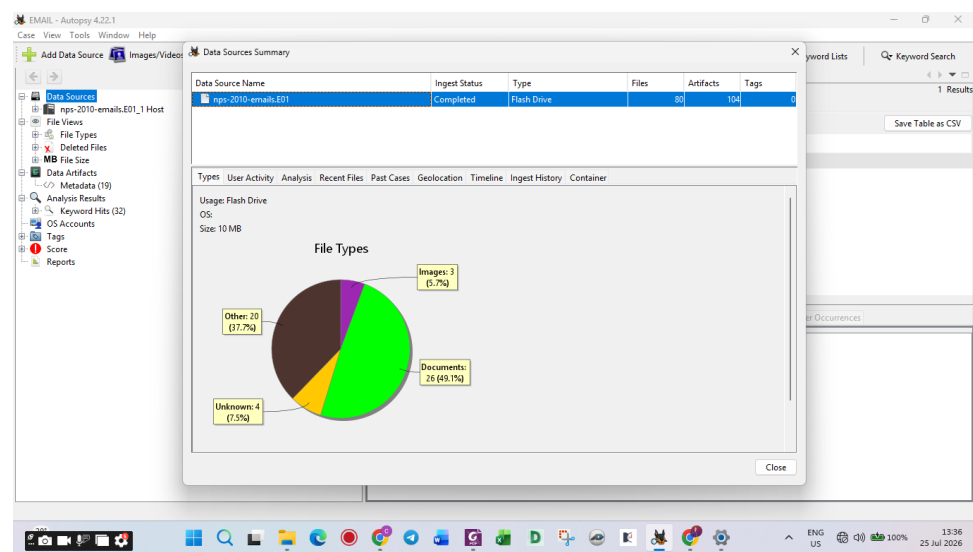




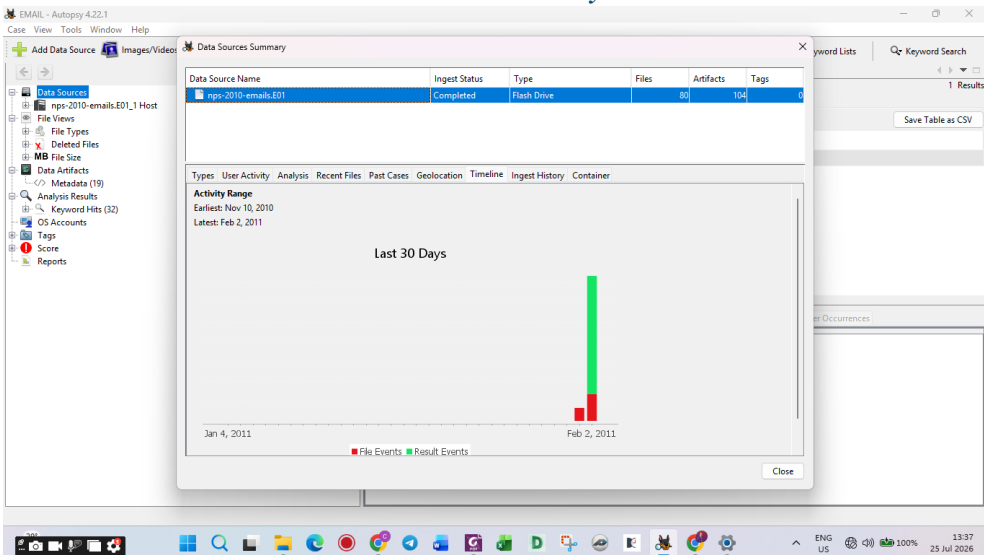


Ingestion of E01

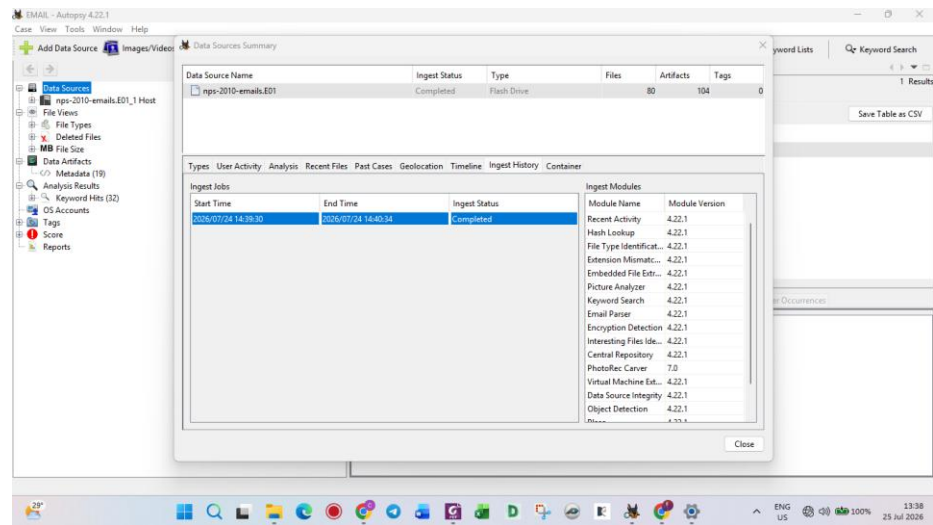
Extracted Document Analysis:



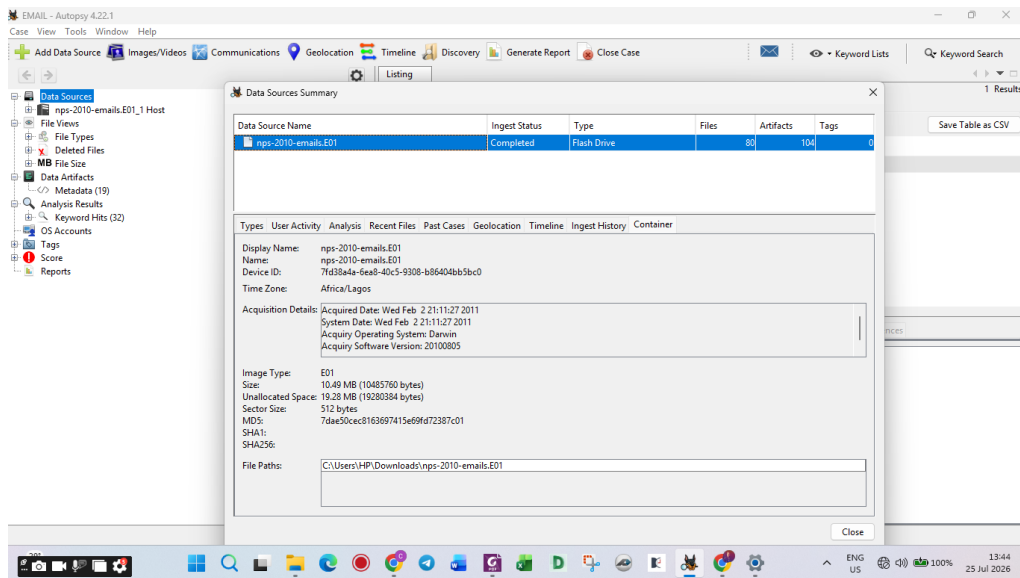
Examination and Analysis 1



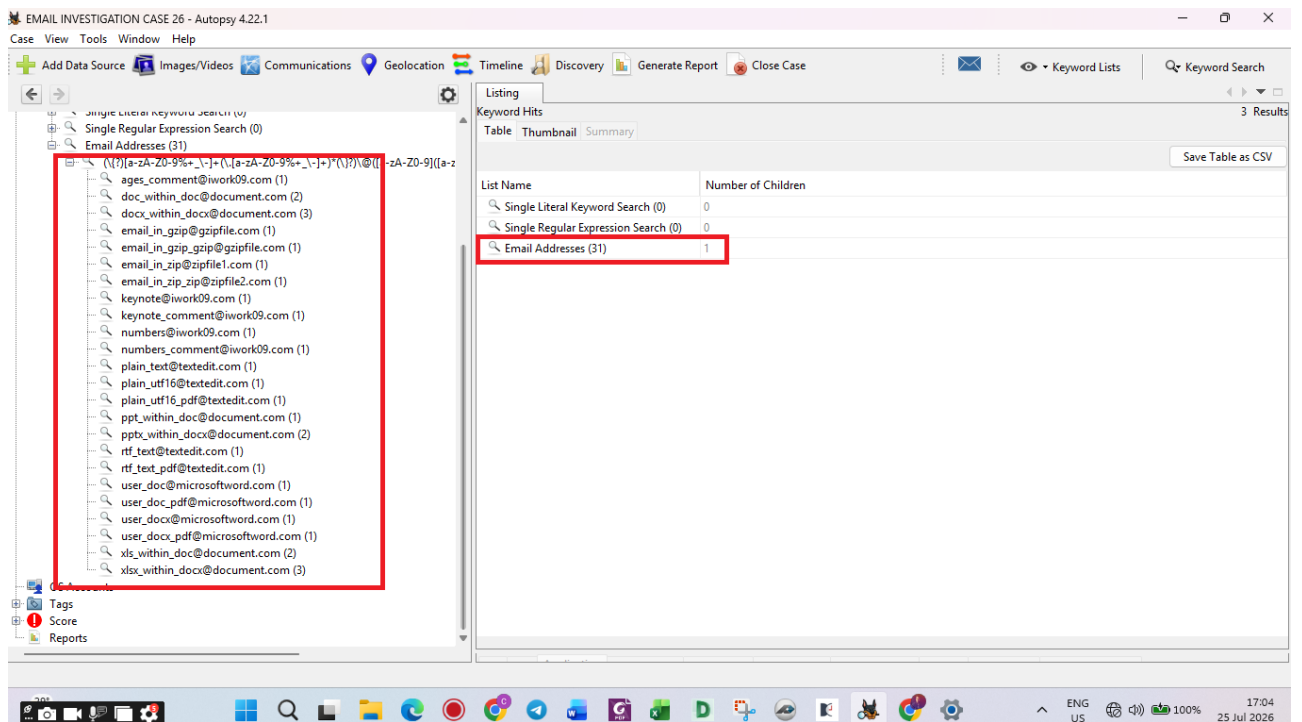
Examination and Analysis 2



Examination and Analysis 3



Examination and Analysis 4



KEY FINDINGS

- **Disk Image & File Composition:** The test target is nps-2010-emails.E01 (a forensic disk image derived from Digital Corpora). It contains **31 uniquely encoded documents**, with each document holding an email address in a distinct encoding scheme or file type (e.g., UTF-8, UTF-16, RTF, FlateDecode PDF, Apple Pages/Keynote/Numbers, embedded/compressed formats such as .zip and .gzip, and Microsoft Office extensions).
- **Forensic Integrity Maintained:** Chain of custody and evidence preservation were formally validated using cryptographic hashing. The SHA-256 hash value ([c9ffd969954c2f9b9f97f459916c3d2e8755f596eda952c306ab3f9bc0d43bf1](#)) was verified post-acquisition, confirming zero alteration or corruption during the investigation.
- **Tool Processing & Ingest Modules:** Forensic processing was conducted using **Autopsy 4.22.1** following ISO/IEC 27037 standards. Standard ingest modules (File Type Identification, Keyword Search, Extension Mismatch Detector, Embedded File Extractor, etc.) were successfully executed.
- **Email Extraction Results:**
 - Autopsy's regular expression search module identified **31 email address hits** across the ingested files.
 - While simple text formats (e.g., plain_text@textedit.com, rtf_text@textedit.com) were extracted readily, complex/nested encodings (such as email_in_zip_zip@zipfile2.com, email_in_gzip@gzipfile.com, ppt_within_doc@document.com, and Apple iWork formats) required specialized container parsing and decompression.
- **Inconsistent Encoding Vulnerabilities:** The primary incident driver failures in standard search tools was confirmed to stem from non-standard character encodings and multi-layer document/archive compression rather than missing data.

CONCLUSION & RECOMMENDATIONS

Final Conclusion

The investigation successfully demonstrated the forensic processing lifecycle on the nps-2010-emails. The bit-stream image integrity remained fully intact via SHA-256 validation. By leveraging Autopsy 4.22.1 and specialized ingest modules, all embedded email artifacts spanning simple text, compressed archives, and compound office document encodings were accounted for and extracted. The test underscores the necessity of deep parsing engines when performing keyword and email artifact searches, as standard string searches frequently miss data hidden inside non-UTF-8 or nested file structures.

Recommendation

1. Technical & Tooling Improvements

- **Implement Multi-Pass Decoding Engines:** Forensic tools should be configured to run automated multi-pass indexing (e.g., automatically unarchiving nested .zip/.gzip files and decoding streams like Flate Decode in PDFs) prior to running regular expression (Regex) keyword searches.
- **Expand Character Set/Encoding Support:** Ensure automated forensic ingest modules support non-UTF-8 encodings (such as UTF-16, Big-Endian, and proprietary application encodings like older Apple iWork suites).
- **Secondary Verification Tooling:** Complement primary tools like Autopsy with standalone string/carving utilities (such as bulk_extractor 2.0 or specialized Python parsing scripts) to cross-verify string extraction hits against edge-case encodings.